

AZURE CLOUD GOVERNANCE PROJECT

Azure Resource Naming & Organisation Standard Rollout

University Cloud Engineering Capstone

Course: Cloud Infrastructure & Enterprise Governance

Scope: Enterprise naming conventions, resource organisation & automated policy enforcement



Azure Resource Naming and Organisation Standard Rollout

GOVERNANCE

S-4-MA

TEAM MEMBERS

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Project Objectives

1



Standardize Resource Naming

Define and enforce a common naming convention for all Azure resources.

2



Improve Resource Discoverability

Make resources easy to identify, search and understand across the organization.

3



Check Naming Compliance

Automatically validate resource names against the organizational naming rules.

4



Identify Non-Compliant Resources

Find resources that do not follow the naming standard and highlight the violations.

5



Provide Recommended Names

Suggest meaningful and compliant names for non-compliant resources.

6



Handle Existing Resources and Urgent Deployments

Provide safe remediation for existing resources and a controlled exception mechanism for urgent deployments.



Azure Resource Naming and Organisation Standard Rollout

Governance Naming Resources

STEP BY STEP PROCESS

- **Define a standard naming convention** for Azure resources so that all teams follow a consistent format.
- **Improve resource discoverability** by making resources easy to identify based on their name, type, environment, and location.
- **Automatically identify non-compliant resources** by checking their names against the defined naming rules.
- **Provide recommended names** for resources that do not follow the organizational standard.
- **Handle existing resources safely**, since Azure resources generally cannot be renamed directly in place.
- **Support urgent deployments** through a controlled exception mechanism without completely bypassing governance
- **Improve overall Azure governance** by making resources easier to organize, manage, and monitor.

Why Azure Resource Naming & Organisation Standards?

1 Consistent Naming

Use a common format across Azure resources.

2 Better Discoverability

Quickly identify and locate resources.

3 Governance & Compliance

Enforce organizational naming policies.

4 Easier Management

Simplify monitoring, auditing and cost management.

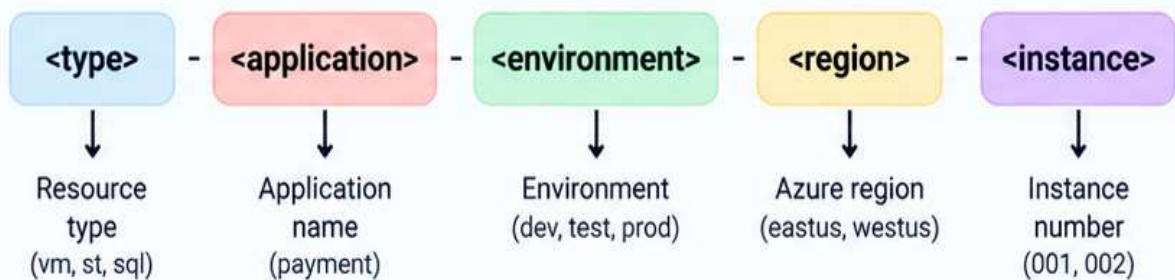
5 Safe Standard Rollout

Handle existing resources and urgent deployments without disrupting operations.

Key Use Cases

1. Naming Convention

Define and enforce a consistent naming format for all Azure resources so that they are easy to identify and manage.



Example:

vm - payment - prod - eastus - 001



Virtual Machine

vm-...



Storage Account

st-...



SQL Database

sql-...



Key Vault

kv-...



App Service

app-...

2. Estate Discoverability

Easily find and understand all Azure resources in the organization using Azure Resource Graph Explorer.



Azure Resource Graph Explorer



Kusto Query (Example)

```
Resources
| project name, type,
  resourceGroup, location, subscriptionId
| order by name asc
```

Discovered Resources

Name	Type	Resource Group	Location	Subscription ID
vm-payment-prod-001	Microsoft.Compute/virtualMachines	rg-payment	eastus	xxxx-xxxx
st-logs-prod-001	Microsoft.Storage/storageAccounts	rg-logs	eastus	xxxx-xxxx
sql-test-dev-001	Microsoft.Sql/servers	rg-sql	westus	xxxx-xxxx
app-portal-prod-001	Microsoft.Web/sites	rg-app	eastus	xxxx-xxxx



Easy to search and identify resources



View type, resource group, location



Helps in compliance and governance

1

Understand Business & Compliance Needs

- Identify organizational goals, regulatory requirements and governance needs.
- Define what needs to be secured and managed in Azure.



2

Implement GRC (Governance, Risk and Compliance)

- Define policies, standards and roles.
- Identify and assess risks.
- Ensure compliance with internal and external regulations.



3

Set Up Azure Organisation Structure

- Create Azure Tenant (root identity).
- Use Management Groups for governance.
- Create Subscriptions for billing and administrative boundaries.



4

Apply Resource Naming Standards

- Define a consistent naming convention (e.g., <env>-<app>-<resource>-<region>).
- Apply naming rules for resources, tags and resource groups.
- Use Azure Policy or templates to enforce naming standards.



5

Rollout Organisation Standards

- Apply common policies, naming standards and tagging across all subscriptions.
- Use automation tools (Azure Policy, ARM templates, etc.).
- Validate and monitor to ensure compliance.



6

Result

- Well-governed, secure and organized Azure environment.
- Consistent resource naming and structure.
- Reduced risks, easier compliance and simplified management.

Advantages

01



Consistent Naming

Provides a standard naming format across all Azure resources.

02



Better Discoverability

Makes resources easy to find, search and understand.

03



Improved Management

Helps in better monitoring, organization and cost management of resources.

04



Compliance and Governance

Ensures resources follow organizational policies and naming rules.

05



Increased Collaboration

Helps different teams follow the same naming standard, reducing confusion.

06



Operational Efficiency

Reduces time spent in identifying resources and minimizes unnecessary costs.

Disadvantages

01



Cannot Rename Existing Resources

Most Azure resources cannot be renamed directly in place, so remediation can be complex.

02



Initial Effort Required

Setting up naming standards and validation processes requires time and planning.

03



Resistance from Teams

Teams may find it challenging to adopt new naming conventions initially.

04



Migration Complexity

For non-compliant resources, creating replacement resources and migrating data can be complex.

05



Risk of Incorrect Naming

If naming rules are not properly followed, it can still lead to inconsistent or incorrect resource names.

06



Handling Urgent Deployments

Strict naming enforcement may delay critical deployments, so a controlled exception mechanism is needed.

Azure Resource Organisation Hierarchy

- 1 Azure Tenant**
Dedicated Microsoft Entra identity and authorization root boundary

- 2 Management Groups**
Policy, RBAC, and governance container across multiple subscriptions

- 3 Subscriptions**
Billing demarcation, quota unit, and administrative boundary

- 4 Resource Groups**
Logical deployment lifecycle container for co-managed resources

- 5 Azure Resources**
Individual instantiated services: VMs, storage accounts, networks

THANK YOU